

GrantSetu: A Verification-Centric Multi-Agent Retrieval-Augmented Generation Architecture for Non-Profit Grant Engineering

Technical Implementation Methodology, Algorithmic Formulations, and Empirical Verification

Tanvi Khadatkar, Jiya Aswani, Deepmalika Das, Ojasvi Maladkar

Project Guide: Dr. Shachi Natu

Department of Information Technology, Thadomal Shahani Engineering College (TSEC), Mumbai, India

Academic Year 2026-2027 | Capstone Project Group 22

ABSTRACT

Non-governmental organizations (NGOs) in emerging economies face severe structural friction when navigating fragmented grant ecosystems, stringent statutory compliance mandates, and labor-intensive proposal writing. While Large Language Models (LLMs) demonstrate remarkable generative capabilities, their deployment in institutional grant engineering is hindered by unconstrained factual hallucinations—particularly with respect to organizational track records, operational vintage, and statutory registration credentials (e.g., NITI Aayog NGO Darpan, FCRA, Sections 12A/80G). In this paper, we present the design and empirical implementation of **GrantSetu**, an end-to-end autonomous, multi-agent Retrieval-Augmented Generation (RAG) system. GrantSetu coordinates a stateful directed acyclic graph (DAG) via LangGraph to execute hybrid multi-stage grant discovery, non-LLM deterministic statutory gating, multi-template modular proposal generation, and algorithmic budget double-counting detection. To eliminate hallucination, we formalize and implement an adapted FActScore verification engine that dynamically decomposes drafted proposals into atomic propositions, validating them through a dual-layer pipeline: deterministic arithmetic/format validation and document-grounded Natural Language Inference (NLI) entailment scoring against an institutional document vault. Empirical benchmarks across real-world Indian non-profit portfolios demonstrate an unconstrained baseline fabrication rate of 41.7%-50.0%, which is systematically mitigated via grounded claim verification and iterative self-correction, supported by a fault-tolerant multi-provider infrastructure across Google Gemini and Groq Llama 3.3-70B.

Index Terms—Multi-Agent Systems, Retrieval-Augmented Generation (RAG), FActScore, Hallucination Mitigation, Reciprocal Rank Fusion, LangGraph, Statutory Compliance Automation, Natural Language Inference.

I. INTRODUCTION AND DOMAIN MOTIVATION

The social impact sector in India comprises over 3.2 million non-governmental organizations (NGOs) operating across vital development domains including education, healthcare, rural livelihoods, and environmental restoration. However, fewer than 10% of these organizations possess the administrative and technical bandwidth required to secure competitive institutional funding. Grant seeking remains severely bottlenecked by two converging pressures: (1) *extreme regulatory fragmentation*, encompassing mandatory registration under the NITI Aayog NGO Darpan portal, tax exemption certifications under Sections 12A/12AB and 80G of the Income Tax Act, and strict foreign funding clearances under the Foreign Contribution Regulation Act (FCRA, as amended in 2020); and (2) *exhaustive proposal drafting lifecycles*, which demand customized thematic alignment, detailed Logical Frameworks (LogFrames), and mathematically rigorous budgets.

While contemporary generative artificial intelligence models offer fluent text generation, off-the-shelf generative models fail catastrophically in institutional grant engineering. Unconstrained LLMs frequently hallucinate quantitative metrics (e.g., claiming 50,000 beneficiaries when actual records state 5,000), invent prior grant affiliations, or fabricate statutory credentials. In institutional grant evaluations, a single fabricated registration number or incompatible vintage calculation results in immediate disqualification and potential blacklisting. GrantSetu resolves this fundamental challenge through a verification-centric architecture that strictly decouples probabilistic language generation from deterministic statutory verification.

II. STATEFUL MULTI-AGENT GRAPH ARCHITECTURE

GrantSetu is architected as a stateful, cyclic Directed Acyclic Graph (DAG) implemented using LangGraph. The system models the grant discovery, statutory filtering, proposal synthesis, and factual verification lifecycle as an immutable state transition system $G = (V, E, S)$, where the global state S is defined as a strictly-typed tuple:

$$S = \blacksquare N_{\text{profile}}, G_{\text{target}}, D_{\text{vault}}, P_{\text{draft}}, C_{\text{claims}}, F_{\text{audit}}, R_{\text{count}} \blacksquare$$

The node set V comprises specialized autonomous functional units executing discrete tasks:

- **Discovery Node (n_{disc})**: Executes multi-stage sparse-dense retrieval over a curated catalog of 150+ verified Indian grant opportunities.
- **Statutory Eligibility Gate (n_{elig})**: A 100% deterministic rules engine (strictly non-LLM) validating NITI Aayog Darpan format, FCRA validity, Sections 12A/80G status, and operational vintage.
- **Proposal Synthesizer (n_{draft})**: Executes modular section-level drafting conditioned on NGO vault context and grant guidelines across three institutional templates.
- **Atomic Claim Extractor ($n_{extract}$)**: Deconstructs generated proposal prose into independent, verifiable atomic factual propositions.
- **Hybrid Verification Node (n_{verify})**: Grades extracted claims via dual-layer deterministic rules and natural language inference against document vault chunks.
- **Revision Router (n_{route})**: Evaluates proposal fabrication rate against threshold $\tau = 0.15$; routes to targeted revision or final export.

III. HYBRID INFORMATION RETRIEVAL AND STATUTORY GROUNDING

To ensure high recall across domain-specific jargon and semantic concepts, GrantSetu implements a multi-stage hybrid retrieval architecture combining sparse linguistic indexing and dense semantic vector search over PostgreSQL utilizing the pgvector extension.

Unstructured grant documents, guidelines, and uploaded NGO filings are segmented into text chunks using recursive boundary-aware windowing. Dense vectors (1024-dimensional) are generated using Google gemini-embedding-001 and indexed using Hierarchical Navigable Small World (HNSW) graphs. In parallel, lexical full-text search is maintained via PostgreSQL tsvector executing English stemming and BM25 term frequency weighting. The individual ranking lists are unified via Reciprocal Rank Fusion (RRF):

$$RRF_Score(d \in D) = \sum_{m \in \{Dense, Sparse\}} [1 / (k + r_m(d))], \quad k = 60$$

where $r_m(d)$ denotes the ordinal rank of document chunk d under retrieval modality m , and smoothing constant $k = 60$ mitigates outlier bias. The top candidates are subsequently processed by a cross-encoder reranker powered by Google Gemini 3.6 Flash, computing calibrated relevance scores and contextual rationales.

IV. DOMAIN-CONSTRAINED MULTI-TEMPLATE PROPOSAL GENERATION

Grant applications exhibit substantial structural divergence depending on the funding body. GrantSetu incorporates three dedicated institutional grant templates:

- 1) **Standard Philanthropic Template**: Optimized for bilateral institutions, global foundations, and multi-year development programs. Emphasizes problem definition, SMART objectives, theory of change, and risk mitigation.
- 2) **Corporate Social Responsibility (CSR) Template**: Formatted specifically to address Section 135 and Schedule VII of the Indian Companies Act, 2013. Mandates clear mapping to eligible CSR sub-sectors (e.g., healthcare, education, gender equality), local geographic stakeholder mapping, and quantifiable ESG impact metrics.
- 3) **Government Grant-in-Aid (GIA) Template**: Tailored to the rigorous administrative conventions of Indian Central and State Ministries (e.g., MWCD, MoSJE, DST). Incorporates Scheme code linkages, NITI Aayog NGO Darpan credential headers, and statutory administrative cost limits.

Algorithmic Budget Integrity & Double-Counting Safeguards

Financial inaccuracies in grant proposals are a primary trigger for audit rejection. GrantSetu integrates an algorithmic budget validator directly into the drafting node. Beyond verifying that line-item sums adhere to total project caps ($\sum b_i \leq C_{max}$), the engine implements a subset-sum heuristic (`_detect_subset_sum_overlap`): Given line-item amounts $B = \{b_1, ..., b_m\}$, the algorithm identifies if $\exists S \subset B$ such that $\sum_{x \in S} x = b_j$ for any $b_j \in B \setminus S$. This heuristic flags when sub-component activities are mistakenly duplicated as separate top-level expenditure heads. Furthermore, the engine programmatically enforces statutory administrative expense capping (e.g., FCRA/CSR norms capping administrative overhead at 5% of total grant disbursement).

V. FACTSCORE-STYLE HYBRID CLAIM VERIFICATION ENGINE

To establish rigorous factual fidelity, GrantSetu adapts the FActScore (Factual Precision in Atomic Claims) framework (Min et al., 2023) for mission-critical non-profit compliance. Rather than evaluating entire generated paragraphs through subjective LLM self-reflection, the engine deconstructs proposal narratives into atomic propositions and subjects them to dual-layer verification.

A. Atomic Proposition Decomposition

Given a generated proposal section text P composed of sentences $\{s_1, s_2, \dots, s_n\}$, an extraction operator E deconstructs each sentence into atomic, independent propositions: $E(s_i) = \{c_{i,1}, c_{i,2}, \dots, c_{i,m}\}$. Each claim c represents a singular, uncompound factual assertion pertaining to organizational credentials, beneficiary statistics, financial turnover, project achievements, or geographic reach.

B. Dual-Layer Verification Operator

Each extracted claim c is evaluated by a composite verification function $V(c)$:

$$V(c) = \{ V_{det}(c) \text{ if } c \in C_{statutory} \cup C_{temporal}; V_{NLI}(c, D_{vault}) \text{ otherwise} \}$$

1) Deterministic Verification Layer (V_{det}): Eliminates model hallucination for legal and temporal assertions. Evaluates: (i) *NGO Darpan format validity* using regex pattern $\wedge[A-Z]\{2\}/(20\d\{2\}\d\{4\})\wedge\d\{7\}\$$; (ii) *Operational vintage arithmetic*: $V_{calc} = Y_{current} - Y_{registration} \geq V_{required}$; (iii) *Statutory status*: confirms 12A/12AB registration, Section 80G tax-exemption approval, and FCRA active validity against institutional records.

2) NLI Entailment Layer (V_{NLI}): For empirical and narrative claims, relevant evidence chunks are retrieved from the NGO Document Vault. A calibrated LLM acting as a Natural Language Inference (NLI) classifier evaluates the premise-hypothesis relationship, categorizing each claim into one of three discrete states: SUPPORTED (premise entails claim with citation chunk ID), UNSUPPORTED (insufficient evidence in vault), or CONTRADICTED (evidence directly refutes claim).

C. Mathematical Formulations of Factuality Metrics

We formalize the quantitative verification metrics governing proposal evaluation:

$$\text{Fabrication Rate: } FR(P) = |\{c \in C(P) : V(c) = UNSUPPORTED\}| / |C(P)|$$

$$\text{Contradiction Rate: } CR(P) = |\{c \in C(P) : V(c) = CONTRADICTED\}| / |C(P)|$$

$$\text{Factuality Score: } FS(P) = |\{c \in C(P) : V(c) = SUPPORTED\}| / |C(P)| = 1 - [FR(P) + CR(P)]$$

Proposals with $FR(P) > \tau$ (set to 0.15) trigger an automated targeted revision node that strips unsupported assertions and synthesizes grounded alternatives.

VI. RESILIENT MULTI-PROVIDER LLM ORCHESTRATION

To ensure continuous system availability during high concurrency and overcome strict provider rate limits (e.g., Google Gemini free-tier 15 RPM/1500 RPD limits), GrantSetu implements an adaptive multi-tier LLM failover architecture. The primary generation layer dynamically rotates across the Google Gemini 3.x Flash series (gemini-3.5-flash, gemini-3.5-flash-lite, gemini-3.6-flash, gemini-3.8-flash, gemini-3.7-flash). Upon encountering rate-limiting exceptions (HTTP 429), gateway timeouts (HTTP 504), or high-demand throttling (HTTP 503), the engine automatically fails over to Groq Cloud hosting Meta Llama 3.3-70B Versatile, providing an auxiliary buffer of 1,000 requests/day at near-zero latency overhead. The failover subsystem maintains an in-memory thread-safe cooldown registry: HTTP 504/503 errors trigger a 300-second backoff, while HTTP 429 quota exhaustion activates a 7200-second cooldown.

VII. EMPIRICAL BENCHMARKS AND QUANTITATIVE EVALUATION

The GrantSetu architecture was evaluated on three verified Indian NGO profiles spanning diverse operational scales: Child Rights and You (CRY; national pan-India operations), Each One Teach One (EOTO; regional education focus), and Pratham Education Foundation (large-scale literacy).

NGO Profile	Framework	Extracted Claims	Supported	Unsupported	Baseline FR (%)	Post-Grounded FR (%)
CRY India	CSR (Sched. VII)	12	7	5	41.7%	0.0%
Each One Teach One	Govt. GIA	10	5	5	50.0%	0.0%
Pratham Foundation	Standard Philanthropic	14	9	5	35.7%	7.1%
Average / Aggregate	All Frameworks	36	21	15	42.5%	2.4%

TABLE I: Empirical verification benchmarks across representative Indian non-profit profiles showing baseline fabrication rates and post-grounding mitigation.

As detailed in Table I, unconstrained generative proposal drafting yielded an average baseline fabrication rate of 42.5%. Unsupported assertions predominantly involved exaggerated beneficiary outreach numbers, specific funding disbursements, and unwarranted future

projections. Following FActScore atomic decomposition and grounded revision routing, the fabrication rate dropped to an aggregate 2.4%, demonstrating near-complete factual fidelity. System reliability was validated through an automated pytest suite containing **24 comprehensive unit tests** executing in 1.62s. Tests confirmed deterministic state transitions, subset-sum duplicate detection, statutory gate barriers, and database proposal revision auto-incrementation.

VIII. CONCLUSION AND FUTURE TRAJECTORY

GrantSetu establishes a rigorous architectural blueprint for deploying generative artificial intelligence in high-stakes public sector and non-profit domains. By enforcing deterministic statutory gating and grounding generative text in verifiable document vault evidence via an adapted FActScore protocol, the system eliminates disqualifying hallucinations while automating proposal authoring. Future research trajectories include integrating continuous semantic benchmarking pipelines via DeepEval and RAGAS, extending tabular OCR parsing for complex financial audit reports, and expanding multi-lingual proposal synthesis across Indian vernacular languages (Hindi, Marathi, Tamil).

REFERENCES

- [1] S. Min, K. Krishna, X. Lyu, M. Lewis, W. Yih, P. Kohli, M. Iyyer, L. Zettlemoyer, and H. Hajishirzi, "FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, Dec. 2023.
- [2] P. Lewis, E. Perez, A. Piktus, et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459-9474, 2020.
- [3] G. V. Cormack, C. L. Clarke, and S. Buettcher, "Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods," in *Proc. ACM SIGIR Conf. Research and Development in Information Retrieval*, pp. 758-759, 2009.
- [4] LangChain AI, "LangGraph: Building Stateful, Multi-Actor Applications with LLMs," *Technical Documentation and Framework Architecture*, 2024.
- [5] Ministry of Corporate Affairs, Government of India, "Companies (Corporate Social Responsibility Policy) Rules, 2014 and Schedule VII Amendments," *The Gazette of India*, 2014-2023.
- [6] Ministry of Home Affairs, Government of India, "The Foreign Contribution (Regulation) Amendment Act, 2020," *Act No. 33 of 2020*, Sep. 2020.
- [7] NITI Aayog, Government of India, "NGO Darpan Guidelines and Unique Identification Portal Specification," *NITI Aayog Official Documentation*, 2021.